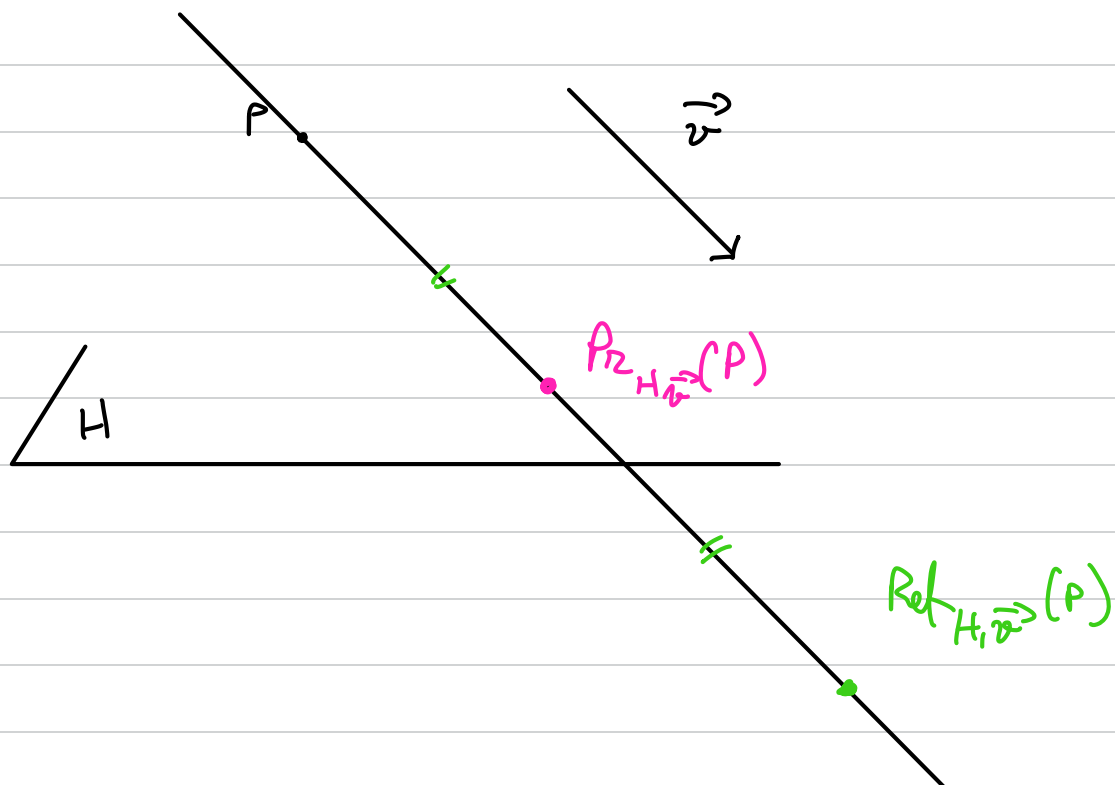


Seminar 6 - Projections & reflections

H - hyperplane

$$H: \alpha_1 x_1 + \dots + \alpha_m x_m + \alpha_{m+1} = 0$$

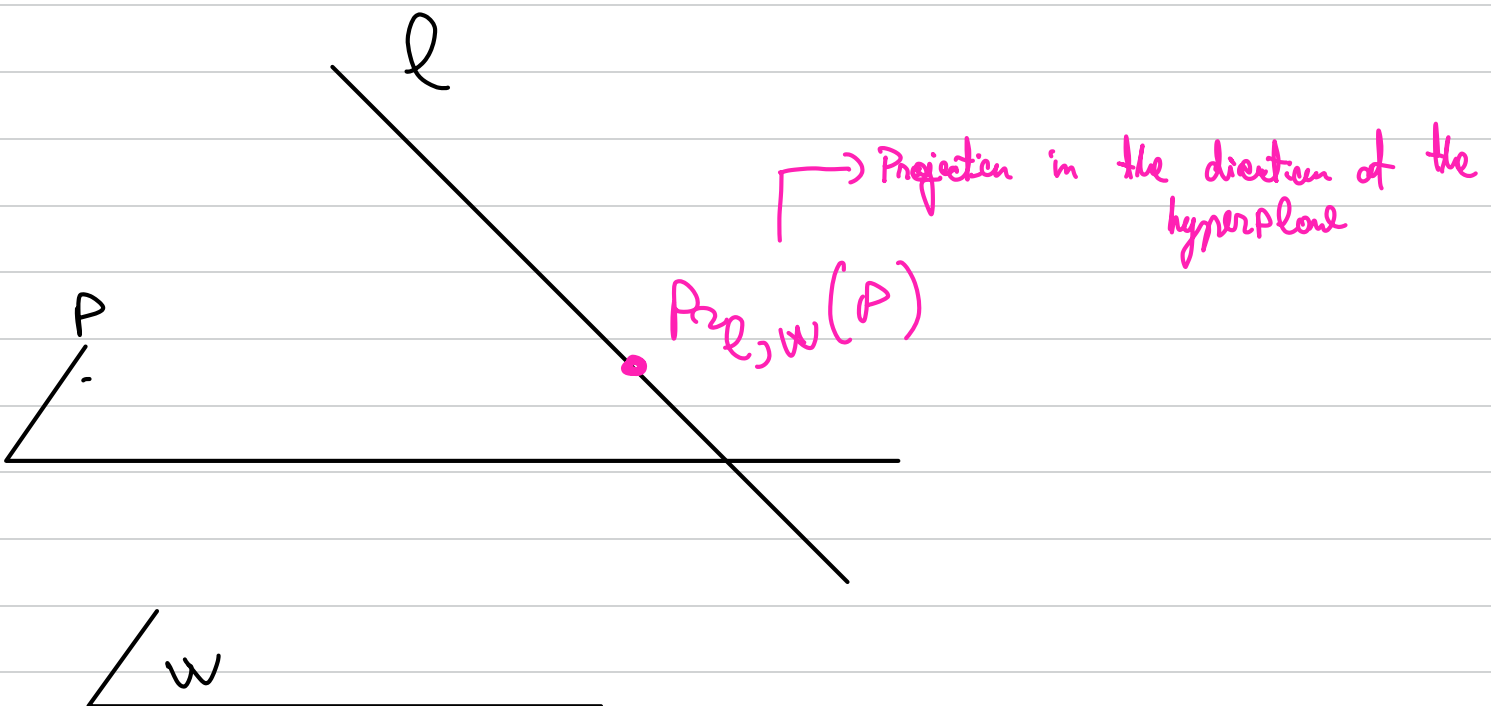
$$\vec{v} \in \mathbb{R}^m$$



$$P_{H, \vec{v}}(P) = \left(I_m - \frac{\vec{v} \cdot \vec{a}^T}{\vec{v}^T \cdot \vec{a}} \right) \cdot P - \frac{\alpha_{m+1}}{\vec{v}^T \cdot \vec{a}} \cdot \vec{v}$$

$\vec{a} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix}$

$$\begin{aligned}
 \text{Proj}_{H,v}(P) &= 2 \text{Proj}_{H,v}(P) - P \\
 &= \left(2 \text{Proj}_m - 2 \frac{v a^T}{v^T a} \right) P - \\
 &\quad \frac{2 \frac{a_{n+1}}{v^T a} \cdot v}{\text{Proj}_m \cdot P} P - \\
 &= \left(\text{Proj}_m - \frac{2 \frac{v a^T}{v^T a}}{\frac{2 \frac{a_{n+1}}{v^T a} \cdot v}{\text{Proj}_m \cdot P}} \right) P -
 \end{aligned}$$



$$w: a_1 x_1 + \dots + a_n x_n = 0$$

$$P_{\ell, \omega}(P) = \frac{v \cdot a^T}{v^T \cdot a} \cdot P + \left(y_m - \frac{v \cdot a^T}{v^T \cdot a} \right) \cdot Q$$

$$v \in D(\ell), Q \in \ell$$

4.2. Determine the orthogonal projection of the line

$$\ell: \begin{cases} 2x - y - 1 = 0 \\ x + y - z + 1 = 0 \end{cases}$$

with the plane

$$\Pi: x + 2y - z = 0$$

(do this by determining the normal fan of the projection)

$$n\text{-normal vector: } \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\ell: \begin{cases} 2x - y - 1 = 0 \\ x + y - z + 1 = 0 \end{cases}$$

$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ 1 & 1 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & -1 & 1 \\ 2 & -1 & 0 & -1 \end{pmatrix} \\ \sim \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & -3 & 2 & -3 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 - \frac{2}{3} & 1 & 1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 1 & -\frac{2}{3} & 1 \end{pmatrix}$$

$$\Leftrightarrow \ell: \begin{cases} x - \frac{1}{3}z = 0 \\ y - \frac{2}{3}z + 1 = 0 \end{cases}$$

$$t = \frac{1}{3}z$$

$$\Leftrightarrow \begin{cases} x = t \\ y = -1 + 2t \end{cases}$$

$$\Leftrightarrow \ell: \begin{cases} x = t \\ y = -1 + 2t \\ z = 3t \end{cases}$$

$$\vec{n}(1, 2, -1)$$

$$Pr_{\pi, \vec{n}}(P) = \left(\frac{1}{\sqrt{3}} - \frac{\vec{n} \cdot \vec{n}^T}{\vec{n}^T \cdot \vec{n}} \right) \cdot P$$

$$n \cdot n^T = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix}$$

$$n^T \cdot n = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = 6$$

$$P_{2 \perp}^{\perp}(P) = \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{6} \cdot \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix} \right) \cdot P$$

$$= \begin{bmatrix} \frac{5}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} \cdot P$$

$$P_{2 \perp}^{\perp}(Q) = \begin{bmatrix} \frac{5}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} \cdot \begin{bmatrix} t \\ -1+2t \\ 3t \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \cdot \begin{bmatrix} t \\ -1+2t \\ 3t \end{bmatrix}$$

... area area, calcolo: tra

7.4. H - hyperplane

v - vector, $v \perp H$

Use the deduced matrix form to show that

$$a) P_{H,v} \circ P_{H,v} = P_{H,v}$$

$$b) \text{Ref}_{H,v} \circ \text{Ref}_{H,v} = \text{Id}$$

$$P_{H,v}(P) = \left(I_n - \frac{v \cdot v^T}{v^T \cdot v} \right) P - \frac{a_{n+1}}{v^T \cdot v} \cdot v$$

We must show that $P_{H,v}(P_{H,v}(P)) = P_{H,v}(P)$

$$\begin{aligned} P_{H,v}(P_{H,v}(P)) &= \left(I_n - \frac{v \cdot v^T}{v^T \cdot v} \right) \cdot \left(\left(I_n - \frac{v \cdot v^T}{v^T \cdot v} \right) P - \frac{a_{n+1}}{v^T \cdot v} \cdot v \right) - \frac{a_{n+1}}{v^T \cdot v} \cdot v \\ &= \left(I_n \left(I_n - \frac{v \cdot v^T}{v^T \cdot v} \right) P - I_n \cdot \frac{a_{n+1}}{v^T \cdot v} \cdot v \right) \end{aligned}$$

$$- \frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \cdot \left(\vec{y}_m - \frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \right) \vec{P} + \frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \cdot \frac{\alpha_{m+1}}{\vec{u}^T \cdot \vec{a}} \cdot \vec{v}$$

$$- \frac{\alpha_{m+1}}{\vec{u}^T \cdot \vec{a}} \cdot \vec{v}$$

$$= \left(\vec{y}_m - \vec{y}_m \cdot \frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \right) \vec{P} - \frac{\alpha_{m+1}}{\vec{u}^T \cdot \vec{a}} \cdot \vec{v}$$

$$- \left(\frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} - \left(\frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \right)^2 \right) \vec{P} + \cancel{\vec{v} \left(\frac{\vec{v} \cdot \vec{a}^T \cdot \alpha_{m+1}}{(\vec{u}^T \cdot \vec{a})^2} \right)} \\ - \cancel{\frac{\alpha_{m+1}}{\vec{u}^T \cdot \vec{a}}} \right)$$

$$= \left(\vec{y}_m - \frac{\vec{v} \cdot \vec{a}^T}{\vec{u}^T \cdot \vec{a}} \right) \vec{P} - \frac{\alpha_{m+1}}{\vec{u}^T \cdot \vec{a}} \cdot \vec{v} \\ - \left(\frac{\vec{v} \cdot \vec{a}^T \cdot \vec{u}^T \cdot \vec{a} - (\vec{v} \cdot \vec{a}^T)^2}{(\vec{u}^T \cdot \vec{a})^2} \right) \vec{P}$$

$$\underbrace{\vec{v} \cdot \vec{a}^T \cdot \vec{v} \cdot \vec{a}^T}_{\text{scalar}} = (\vec{a}^T \cdot \vec{v}) (\vec{v} \cdot \vec{a}^T)$$

$$\vec{a}^T \vec{u} = (\vec{a}^T \cdot \vec{u})^T = \vec{u}^T \cdot \vec{a}$$

$$\underbrace{v \cdot a^T \cdot v}_{\text{scalar}} = a^T \cdot v \cdot v = (v^T \cdot a) \cdot v$$

...

$$P_{H_v}^2(P) = \left(I_n - \frac{2 \cdot v \cdot a^T}{v^T \cdot a} + \frac{1}{v^T \cdot a} \cdot v \cdot a^T \right) \cdot P$$

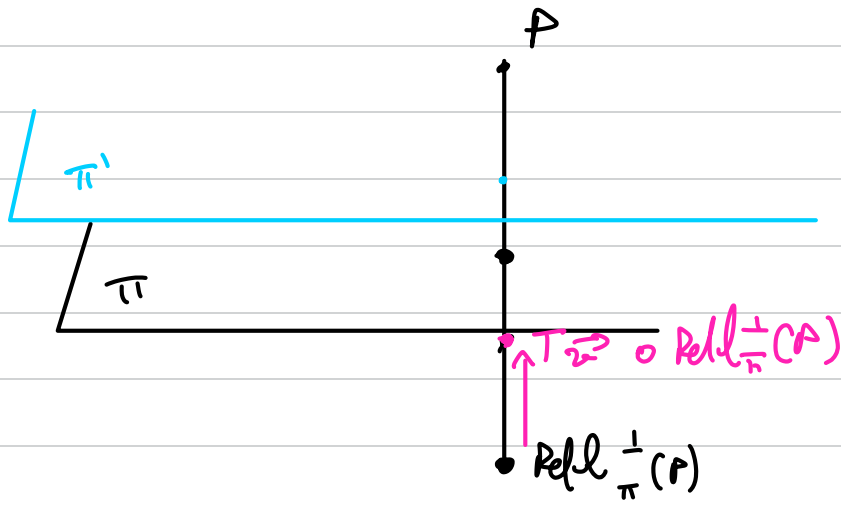
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7.6. Π -plane with normal vector $\vec{n}$ and equation:

$$\Pi: \langle \vec{n}, \vec{x} \rangle = c$$

$$\Leftrightarrow n_1 x_1 + n_2 x_2 + n_3 x_3 - c = 0$$

Show that the composition of the orth. reflection in Π followed by a translation with $\vec{v} \parallel \vec{n}$ is a reflection in the plane Π' and deduce the plane's eqn.



$$\text{Ref}_{\frac{1}{\pi}}^{\perp}(P) = 2 \left(\frac{\vec{m} \cdot \vec{m}^T}{\vec{m}^T \cdot \vec{m}} \right) P + \frac{2c}{\vec{m}^T \cdot \vec{m}} \vec{m}$$

$$\vec{v} \parallel \vec{m} \Rightarrow \exists k \text{ s.t. } \vec{v} = k \vec{m}$$

$$T_2 \circ \text{Ref}_{\frac{1}{\pi}}^{\perp}(P) = \left(\frac{\vec{m} \cdot \vec{m}^T}{\vec{m}^T \cdot \vec{m}} \right) \cdot P + \frac{2c}{\vec{m}^T \cdot \vec{m}} \cdot \vec{m} + k \cdot \vec{m}$$

$$= \left(\frac{\vec{m} \cdot \vec{m}^T}{\vec{m}^T \cdot \vec{m}} \right) \cdot P + \frac{2c + k \vec{m}^T \cdot \vec{m}}{\vec{m}^T \cdot \vec{m}} \cdot \vec{m}$$

$$= \text{Ref}_{\pi'}^{\perp}(P), \text{ where } \pi' = \langle \vec{m}, \vec{x} \rangle = c'$$

$$\pi' : \langle \vec{m}, \vec{x} \rangle = c'$$